ALLIANCE UNIVERSITY · BCA (AI/ML) · DBMS ASSIGNMENT SERIES

07

ASSIGNMENT 7

Sample Data Insertion

Online Food Delivery System

SUBMITTED BY
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PROGRAMME
BCA (AI/ML)

INSTITUTION
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IN THIS ASSIGNMENT

01 Seed records

02 Respect order

03 Cover statuses

04 Check counts

A consistent relational design carried forward from Assignments 1 and 2.

1 Objective and Brief Coverage

A schema is only useful once it holds data. This assignment loads **84 fictional rows** into all eight tables so that the queries in Assignments 8 and 9 return meaningful, varied results. The data is designed to exercise every status value, every optional relationship and every constraint added in Assignment 6.

84	8	12	0
rows inserted	tables populated	orders, 6 statuses	constraint violations

	Brief requirement	Where it is covered
✓	**Insert realistic sample data into all tables**	Section 4: every table printed in full
✓	**Include sufficient records for a real-world scenario**	Section 3: 7 customers to 21 order lines; every status covered
✓	**Data follows constraints and relationships**	Section 2 (insert order) and Section 6 (validation queries, all pass)

Safety note

All names, **example.com** e-mail addresses and phone numbers are synthetic. Password_Hash values are placeholders, not real hashes or credentials. Never load this data into a real service.

2 Insert Order and Referential Integrity

Foreign keys make the database reject a child row whose parent does not exist yet, so data must be loaded **parents first, children last**.

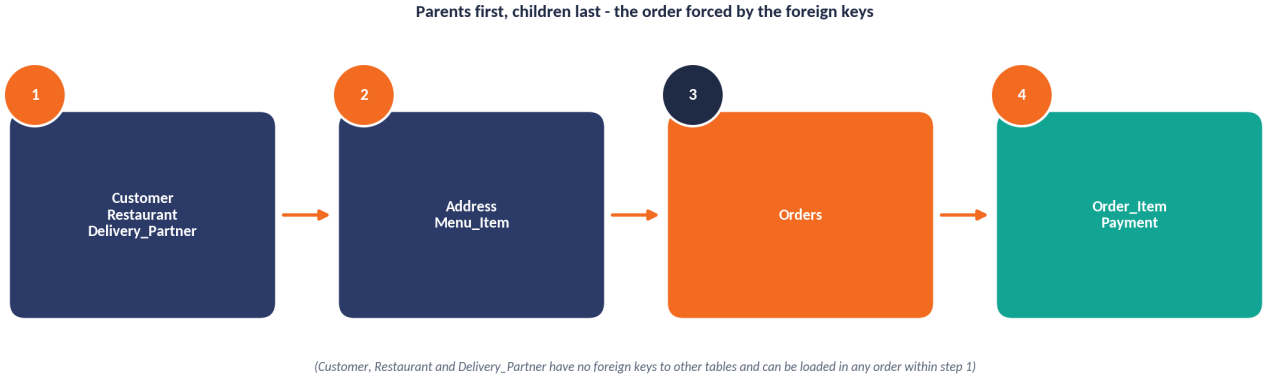


Figure 1. Dependency-driven load order used by the script.

- **Step 1.** Customer, Restaurant, Delivery_Partner have no foreign keys to other tables.
- **Step 2.** Address needs Customer; Menu_Item needs Restaurant.
- **Step 3.** Orders needs Customer, Restaurant, Address (as a pair with Customer) and, optionally, Delivery_Partner.
- **Step 4.** Order_Item needs Orders and Menu_Item; Payment needs Orders.

Explicit IDs

The script supplies explicit primary-key values so that the foreign-key references in later inserts are predictable and the documented outputs are reproducible. After such an insert, AUTO_INCREMENT continues from the highest existing ID, so later inserts without an ID still work.

3 Sample Data Volume and Coverage

Table	Rows	What the data demonstrates
Customer	7	Unique e-mail and phone for each account
Address	8	Customer 1 has two addresses; every order uses an address owned by its customer
Restaurant	5	Ratings 4.0 to 4.5; three cities (Bengaluru, Hosur, Chennai)
Menu_Item	15	Three items per restaurant; prices Rs 40 to Rs 329
Delivery_Partner	4	Available, Busy and Offline statuses
Orders	12	Delivered, Preparing, Out for Delivery, Confirmed, Placed and Cancelled; two orders have no partner yet
Order_Item	21	Single and multi-item orders; quantities 1 and 2; price snapshots
Payment	12	Paid, Pending, Failed, Refunded; Cash, Card, UPI, Wallet

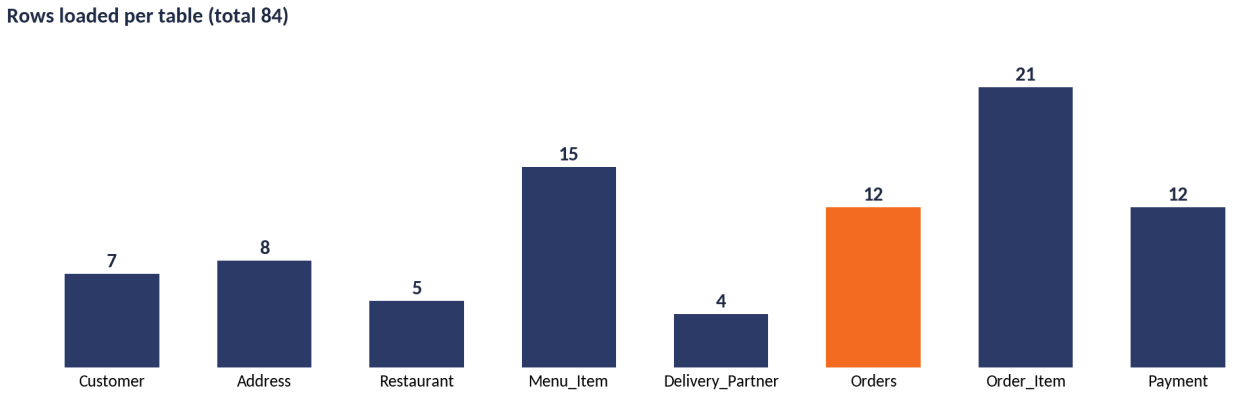


Figure 2. Rows per table; Order_Item is the largest because each order has one or two lines.

Status coverage

Every value allowed by the Assignment 6 CHECK constraints for order and delivery status appears at least once, so later reports can be tested on all of them.

Order status	n	Payment status	n	Payment method	n	Partner status	n
Delivered	6	Paid	6	UPI	6	Available	1
Placed	2	Pending	4	Card	3	Busy	2
Cancelled	1	Failed	1	Cash	2	Offline	1
Confirmed	1	Refunded	1	Wallet	1		
Out for Delivery	1						
Preparing	1						

4 The Complete Dataset

Every row loaded into every table is listed below, in insertion order.

Customer (7 rows)

Customer_ID	Name	Email	Phone	Password_Hash
1	Asha Menon	asha@example.com	9000000001	demo_hash_customer_01
2	Rahul Nair	rahul@example.com	9000000002	demo_hash_customer_02
3	Priya Sharma	priya@example.com	9000000003	demo_hash_customer_03
4	Arjun Reddy	arjun@example.com	9000000004	demo_hash_customer_04
5	Meera Iyer	meera@example.com	9000000005	demo_hash_customer_05
6	Kabir Das	kabir@example.com	9000000006	demo_hash_customer_06
7	Nisha Rao	nisha@example.com	9000000007	demo_hash_customer_07

Address (8 rows)

Address_ID	Customer_ID	House_No	Street	City	Pincode
1	1	12A	Lake View Road	Bengaluru	560001
2	1	B-204	Green Park Apartments	Bengaluru	560034
3	2	44	Temple Street	Hosur	635109
4	3	7C	Garden Main Road	Bengaluru	560076
5	4	18	Market Road	Hosur	635110
6	5	21/3	River View Street	Chennai	600028
7	6	5A	College Road	Hosur	635126
8	7	302	Sunrise Residency	Bengaluru	560068

Restaurant (5 rows)

Restaurant_ID	Restaurant_Name	Phone	Address	Rating
1	Spice Route Kitchen	9100000101	Indiranagar, Bengaluru	4.50
2	Green Bowl Cafe	9100000102	Anna Nagar, Hosur	4.20
3	Urban Pizza Co.	9100000103	Koramangala, Bengaluru	4.40
4	Dosa Junction	9100000104	Railway Station Road, Hosur	4.10
5	Wok Express	9100000105	Adyar, Chennai	4

Delivery_Partner (4 rows)

Delivery_ID	Name	Phone	Vehicle_Number	Status
1	Rohan Singh	9200000101	KA05AB1234	Busy
2	Kavya Patel	9200000102	KA03CD2345	Available
3	Imran Khan	9200000103	TN70EF3456	Busy
4	Sneha Rao	9200000104	KA51GH4567	Offline

Menu_Item (15 rows)

Item_ID	Restaurant_ID	Item_Name	Description	Price	Availability
1	1	Chicken Biryani	Fragrant rice with spices and chicken	220	TRUE
2	1	Paneer Biryani	Spiced paneer rice	180	TRUE
3	1	Vegetable Meals	South Indian vegetarian meal	150	TRUE
4	2	Veg Buddha Bowl	Rice, vegetables and house dressing	160	TRUE
5	2	Paneer Wrap	Paneer and salad in a flatbread wrap	130	TRUE
6	2	Lemon Iced Tea	Fresh lemon iced tea	70	TRUE

Item_ID	Restaurant_ID	Item_Name	Description	Price	Availability
7	3	Margherita Pizza	Tomato, mozzarella and basil	249	TRUE
8	3	Farmhouse Pizza	Vegetables, cheese and tomato sauce	329	TRUE
9	3	Garlic Bread	Toasted garlic bread	119	TRUE
10	4	Masala Dosa	Crisp dosa with potato masala	90	TRUE
11	4	Ghee Roast Dosa	Dosa roasted with ghee	140	TRUE
12	4	Filter Coffee	Traditional filter coffee	40	TRUE
13	5	Veg Hakka Noodles	Stir-fried noodles with vegetables	180	TRUE
14	5	Chilli Paneer	Paneer tossed in Indo-Chinese sauce	220	TRUE
15	5	Spring Rolls	Crisp vegetable spring rolls	120	TRUE

Orders (12 rows)

Order_ID	Customer_ID	Restaurant_ID	Address_ID	Delivery_ID	Order_Date	Order_Status
1	1	1	1	1	2026-10-21 12:15:00	Delivered
2	2	2	3	2	2026-10-21 19:05:00	Delivered
3	3	3	4	3	2026-10-22 13:20:00	Delivered
4	1	4	2	1	2026-10-22 20:00:00	Delivered
5	4	5	5	4	2026-10-23 12:00:00	Out for Delivery
6	5	1	6	2	2026-10-23 13:30:00	Preparing
7	6	3	7	NULL	2026-10-23 19:00:00	Placed
8	2	4	3	3	2026-10-24 08:45:00	Delivered
9	7	2	8	4	2026-10-24 12:00:00	Confirmed
10	3	5	4	NULL	2026-10-24 12:30:00	Placed
11	4	3	5	1	2026-10-20 20:00:00	Delivered
12	5	1	6	2	2026-10-20 12:10:00	Cancelled

Order_Item (21 rows)

Order_ID	Item_ID	Quantity	Unit_Price
1	1	2	220
1	3	1	150
2	4	1	160
2	6	2	70
3	7	1	249
3	9	1	119
4	10	2	90
4	12	2	40
5	13	2	180
5	15	1	120
6	2	1	180
6	1	1	220
7	8	1	329
8	11	1	140
8	12	1	40

Order_ID	Item_ID	Quantity	Unit_Price
9	5	2	130
9	4	1	160
10	14	1	220
10	13	1	180
11	7	2	249
12	3	1	150

Payment (12 rows)

Payment_ID	Order_ID	Payment_Method	Payment_Status	Payment_Date
1	1	UPI	Paid	2026-10-21 12:18:00
2	2	Card	Paid	2026-10-21 19:08:00
3	3	UPI	Paid	2026-10-22 13:23:00
4	4	Cash	Paid	2026-10-22 20:35:00
5	5	UPI	Pending	NULL
6	6	Card	Pending	NULL
7	7	UPI	Pending	NULL
8	8	Cash	Paid	2026-10-24 09:10:00
9	9	Wallet	Pending	NULL
10	10	UPI	Failed	2026-10-24 12:32:00
11	11	Card	Paid	2026-10-20 20:04:00
12	12	UPI	Refunded	2026-10-20 12:40:00

5 Example INSERT Statements

The script uses explicit column lists and multi-row VALUES. Explicit column lists keep the statement valid if columns are later added or reordered.

```
INSERT INTO Customer (Customer_ID, Name, Email, Phone, Password_Hash) VALUES
(1, 'Asha Menon', 'asha@example.com', '9000000001', 'demo_hash_customer_01'),
(2, 'Rahul Nair', 'rahul@example.com', '9000000002', 'demo_hash_customer_02');
-- child rows come after their parents exist
INSERT INTO Order_Item (Order_ID, Item_ID, Quantity, Unit_Price) VALUES
(1, 1, 2, 220.00),   -- 2 x Chicken Biryani at the price on the day
(1, 3, 1, 150.00);   -- 1 x Vegetable Meals
-- nullable foreign key: order not yet assigned to a delivery partner
INSERT INTO Orders (Order_ID, Customer_ID, Restaurant_ID, Address_ID,
                    Delivery_ID, Order_Date, Order_Status)
VALUES (7, 6, 3, 7, NULL, '2026-10-23 19:00:00', 'Placed');
```

6 Validation and Expected Output

Row-count check from the setup script, with the value expected and the value found:

Table_Name	Expected	Found	Result
Customer	7	7	Match
Address	8	8	Match
Restaurant	5	5	Match
Menu_Item	15	15	Match

Table_Name	Expected	Found	Result
Delivery_Partner	4	4	Match
Orders	12	12	Match
Order_Item	21	21	Match
Payment	12	12	Match

Referential and business-rule spot checks (each should return 0 rows):

Check	Rows found	Result
Orders whose address belongs to another customer	0	Pass
Order lines whose menu item is from a different restaurant than the order	0	Pass
Order lines priced differently from the current menu price	0	Pass
Paid or Refunded payments without a payment date	0	Pass
Orders without any order line	0	Pass
Delivered orders whose payment is not Paid	0	Pass

Demonstration queries

Latest five orders

```
SELECT Order_ID, Customer_ID, Restaurant_ID, Order_Status
FROM Orders
ORDER BY Order_Date DESC
LIMIT 5;
```

OUTPUT 5 rows

Order_ID	Customer_ID	Restaurant_ID	Order_Status
10	3	5	Placed
9	7	2	Confirmed
8	2	4	Delivered
7	6	3	Placed
6	5	1	Preparing

Line subtotal per non-cancelled order

```
SELECT o.Order_ID, SUM(oi.Quantity * oi.Unit_Price) AS Line_Subtotal
FROM Orders o
JOIN Order_Item oi ON oi.Order_ID = o.Order_ID
WHERE o.Order_Status <> 'Cancelled'
GROUP BY o.Order_ID
ORDER BY o.Order_ID;
```

OUTPUT 11 rows

Order_ID	Line_Subtotal
1	590
2	300
3	368
4	260
5	480
6	400

Order_ID	Line_Subtotal
7	329
8	180
9	420
10	400
11	498

Reading the validation

Counts match, no order uses another customer's address, and no order line contradicts its restaurant. The only deliberate differences are: some orders are unpaid or not yet delivered, which is what makes the Assignment 8 and 9 reports interesting. Sample Unit_Price values equal the current menu prices because the menu has not changed since these orders were placed.

7 Troubleshooting Failed Inserts

MySQL error	Typical cause here	Fix
1452 Cannot add or update a child row	A parent row does not exist yet (e.g. inserting Orders before Address)	Load parents first (Figure 1)
1062 Duplicate entry	Re-running the script on a populated database, or a repeated e-mail / phone	Run only on an empty schema; use unique test values
3819 Check constraint is violated	Out-of-range rating, zero quantity, or Paid without a date	Correct the value to satisfy the Assignment 6 rule
1364 Field has no default value	A NOT NULL column without a default was omitted	Supply the column or add a DEFAULT

8 Conclusion

The dataset populates every table, respects every key and constraint, and contains enough variety (all statuses, optional delivery assignment, mixed payment methods, multi-item orders) to support meaningful queries. It forms the common foundation for Assignments 8 and 9.

Execution note: the counts and checks above were computed by loading the exact INSERT statements from the setup script into a local SQLite validation harness (no MySQL server was available while preparing this document). Run the MySQL scripts in Workbench for native result-grid screenshots.